Master Thesis Presentation

Good afternoon. My name is Heykel Khadhraoui, and today I will present my master's thesis on real-time impedance control for a seven-degree-of-freedom collaborative robot.

The intended application is surface grinding with a rectangular tool. My work focuses on contact entry: how the tool can rotate towards alignment with a planar surface through compliant contact.

I will begin with the problem that motivates this behaviour.

Motivation

The illustration shows the basic problem. When a rectangular tool approaches a surface with a small tilt, it touches along an edge or at a corner before the whole face makes contact. Small differences in surface placement, tool mounting, and robot pose can produce this situation.

A rigid pose controller tries to maintain the commanded orientation. It therefore resists the corrective rotation that contact could generate, and the contact loads can increase.

The idea in this thesis is to maintain the normal press while allowing the tool to rotate through its interaction with the surface. Here, passive alignment means that contact produces the rotation, rather than an additional prescribed alignment trajectory.

The main question is: how do Cartesian impedance settings affect passive alignment? These settings determine how the controller responds to position and orientation errors, and therefore how readily contact can rotate the tool.

I will first explain the controller, then show how its settings affect the measured contact response.

Overview

The presentation is organised into six chapters. The introduction establishes the contact-alignment problem. Controller design begins with Cartesian pose and covers Cartesian impedance, the real-time loop, surface-relative compliance, centre-of-compliance coupling, and null-space control.

Experimental methodology introduces the contact and null-space experiments and defines the angular quantities. Validation then checks the plausibility and consistency of the controller quantities before the results.

Experimental results proceed from the contact baseline through stiffness and centre-of-compliance variations, followed by null-space damping, conditioning, and Cartesian position retention. The final chapter presents the conclusions and future work. Both video demonstrations remain together in the backup slides.

I will first define the Cartesian pose, then explain how the controller corrects it.

Cartesian pose

Cartesian pose describes both the position and the orientation of the end effector, abbreviated EE, relative to the robot base frame, used here as the reference frame. In three-dimensional space, the end effector's position and orientation together have six degrees of freedom: three for translation and three for rotation.

Both illustrations show the same solid three-dimensional gripper, with its fingers pointing downward and its front turned towards the right about the reference z axis. The marked EE origin lies between the fingers. In the left illustration, five circular joint markers and their connecting links form a schematic chain from the base-frame origin to the gripper. This illustrates how the joint configuration determines the end-effector pose. The curved arrows in the right illustration indicate positive rotations according to the right-hand rule.

The left illustration shows the position vector p. Its three coordinates, x, y, and z, locate the end-effector frame origin in this reference base frame. This position depends on the configuration of all seven robot joint angles through forward kinematics, written $p = p(q)$. The vector q collects those seven joint angles. Orientation also depends on the joint configuration.

The right illustration shows the three rotational degrees of freedom using roll ϕ, pitch θ, and yaw ψ. These describe rotations about the x, y, and z axes. For combined rotations, their order matters. The convention illustrated here is $R = R_z(\psi)R_y(\theta)R_x(\phi)$.

The controller represents orientation with a rotation matrix R. The roll angle on this introductory slide is distinct from the axis–angle error angle used on the next slide.

Position tells us where the end effector is. Orientation tells us how the tool is rotated. The next slide shows how position is corrected through force and orientation through moment.

Cartesian impedance controller

The Cartesian impedance controller corrects position through force and orientation through moment. The first two bullet headings introduce position correction by force and orientation correction by moment. Each equation group is placed directly below its heading.

In the left column, the position error is the desired position minus the measured end-effector position:

$$\begin{aligned} e_p &= p_d - p_{\mathrm{EE}}, \\ f &= K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}}). \end{aligned}$$

The force equation below the position error shows the translational spring and damper contributions.

In the right column, the rotational error describes the correction from measured to desired orientation. Its axis is u, its angle is ϕ, and the error vector is $e_R = \phi u$:

$$\begin{aligned} e_R &= \phi u, \qquad R_d R_{\mathrm{EE}}^\top = R(u, \phi), \\ m &= K_R e_R + D_R(\omega_d - \omega_{\mathrm{EE}}). \end{aligned}$$

The moment equation below the rotational error shows the rotational spring and damper contributions.

Under the third heading, Combined Cartesian wrench, the matrix equation combines these force and moment equations into the six-component wrench F. The translational stiffness K_p and rotational stiffness K_R form the diagonal blocks of the six-by-six Cartesian stiffness matrix K_c. The underbrace identifies the complete matrix, rather than one of its blocks. The damping blocks form D_c in the same way.

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{\mathrm{EE}} \\ \omega_d - \omega_{\mathrm{EE}} \end{bmatrix}.$$

Each stiffness or damping block is three by three, and each zero denotes a three-by-three zero block. In this decoupled form, position error generates force and orientation error generates moment. The zero off-diagonal blocks mean that position error does not add a moment and orientation error does not add a force. The same separation applies to the linear and angular velocity differences through the damping matrix. This is translation–rotation decoupling. It does not require every three-by-three block to be diagonal in the chosen coordinate frame.

The subscripts p and R distinguish translation and rotation. The matrices K and D are stiffness and damping. The subscript d denotes the desired reference, and EE denotes the measured end-effector state. Linear velocity is $\dot{p}$, and angular velocity is ω.

The next slide shows how the commanded Cartesian wrench is applied to the robot.

Real-time control

The loop shows how the controller closes the feedback path. It calculates the pose error from the desired and measured tool pose, generates the impedance wrench, and maps that wrench to joint torques. The robot moves, and its measured pose returns to the next update.

The equation gives the torque mapping. F collects the commanded force and moment. The Jacobian transpose maps this Cartesian wrench to the Cartesian task's joint-torque contribution.

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \qquad \tau_{\text{cart}} = J^{\top}(q)F.$$

Here, q denotes the measured joint positions. The geometric Jacobian, J, depends on this configuration and relates joint motion to Cartesian motion. Its transpose maps the six wrench components to seven joint-torque components.

I implemented this feedback controller in C++ using libfranka. Its nominal torque-control frequency is one kilohertz, or one update every millisecond.

The next slide explains how the compliant behaviour is chosen relative to the surface.

Surface-relative compliance

These directions are defined relative to the configured surface. The arrow n_s is the surface normal: it is perpendicular to the plane and points outward. Pressing towards the surface therefore follows the opposite direction.

The other two arrows, t_1 and t_2, are perpendicular tangent directions lying in the plane. Together, these three directions form the surface frame.

For translation, I use softer behaviour along the normal to allow compliant pressing. Along the two tangents, higher stiffness helps retain the tool's position on the surface.

For rotation, low stiffness about the two tangents allows rotations that bring the tool face towards alignment with the plane. Higher stiffness about the normal resists rotation in the plane and maintains the tool's in-plane orientation.

This distinction is important: translating along a tangent and rotating about that tangent are different motions. The controller can assign a different stiffness to each.

Besides choosing these directions, I can also shift the point about which the impedance is defined.

Centre of compliance (CoC)

The centre of compliance is a virtual reference point for the impedance. Its position is p_c, and r_c is the displacement from the tool centre point to this virtual centre.

The moment equation on the right gives the commanded moment at the TCP. It combines the rotational spring–damper moment, m_R, with the additional contribution from the displaced centre.

$$m = m_R + r_c \times f.$$

Here, f is the commanded force. The two sketches below isolate its normal component, f_n. The upward arrows n_s indicate the surface normal. The sketches show the same tool tilt and normal press, with the compliance centre on opposite sides of the TCP.

Reversing the tangential displacement reverses the additional moment. For a given entry tilt, one side can support the aligning rotation, while the opposite side can oppose it.

At the TCP, r_c is zero, so the additional moment vanishes. The ordinary rotational spring and damper remain active.

The selected CoC defines where the impedance is specified. The added lever contribution is a TCP moment. Expressing the same wrench at the CoC gives $m_c = m - r_c \times f = m_R$. At the TCP, $r_c = 0$ removes this added directional contribution without switching off rotational impedance.

The next slide shows how this geometric point shift transforms the stiffness and damping matrices.

Translation–rotation coupling

The preceding sketches introduced the centre of compliance and its displacement from the TCP. The point-shift transformation A maps the local Cartesian pose error at the TCP to the corresponding error at the centre of compliance:

$$A = \mathrm{Ad}(r_c), \qquad e_c = Ae.$$

This is a local small-angle model. The stiffness and damping matrices are specified independently for translation and rotation at the CoC, with all components expressed in the robot base frame.

The two columns show how the same point shift transforms stiffness and damping to the TCP:

$$K_{\mathrm{TCP}} = A^\top K_c A,$$
$$D_{\mathrm{TCP}} = A^\top D_c A.$$

The transformed matrices generally contain off-diagonal terms when the CoC displacement is non-zero. Translation and rotation are then coupled: translational error and velocity can contribute to the commanded moment, while rotational error and velocity can contribute to the commanded force.

This is the matrix form of the moment contribution illustrated on the preceding slide. At the TCP, r_c is zero, the point shift becomes the identity, and the additional coupling vanishes. The ordinary translational and rotational spring-damper laws remain active.

With the Cartesian controller defined, I will now explain its null-space controller.

Null-space controller

The Redundancy heading describes the available internal freedom. The robot has seven joints, while the Cartesian task specifies six end-effector degrees of freedom: three for position and three for orientation. When the Jacobian has full row rank, one independent null-space direction remains, as shown below the left-hand explanation.

$$\dim(\ker J) = 7 - 6 = 1.$$

Cartesian task preservation describes the effect of coordinated null-space motion in the right-hand column. Its joint velocity produces zero instantaneous end-effector translation and rotation.

$$J(q)\dot{q}_{\text{null}} = 0.$$

Several joints generally move together. This is an instantaneous kinematic relation. The experiment checks the finite position error of the implemented torque controller.

The lower columns distinguish the two controller functions. The damping torque, τ_d, opposes redundant joint velocity and suppresses unwanted joint motion. The conditioning torque, τ_σ, seeks a locally better-conditioned robot configuration and can initiate motion from rest. It is an active configuration objective, so its role differs from damping.

A singularity is a configuration where the Jacobian loses rank. The conditioning objective steers towards better local conditioning, but it does not guarantee avoidance of every singular configuration.

I will now introduce the two experiments used to evaluate the controller.

Contact experiment

The contact experiment uses a rectangular tool measuring forty by one hundred and twenty millimetres against a planar surface. The process sequence shown is tool orientation, surface approach, contact establishment, and surface grinding. The experiments presented here evaluate the five-second Contact Establishment phase.

During approach, translational stiffness is one hundred and fifty newtons per metre in each surface direction. Rotational stiffness about the two tangents is one hundred and eighty newton metres per radian. During Contact Establishment, the settings shown on the left provide compliant normal pressing and soft rotations about the surface tangents.

$$K_{p,n} = 350\,\mathrm{N/m}, \qquad K_{R,t_1} = K_{R,t_2} = 5\,\mathrm{N\,m/rad}.$$

$$K_{p,t_1} = K_{p,t_2} = 2000\,\mathrm{N/m}.$$

$$K_{R,n} = 50\,\mathrm{N\,m/rad}.$$

Normal and tangential translation are specified separately from rotations. Rotational stiffness about the surface normal is fifty newton metres per radian. These are the common contact settings before applying the CoC transformation. Normal translation is softer than tangential translation, and the tangent rotations are much softer than during approach.

The selected tool-point reference advances towards the surface. Contact restricts translation, so the resulting position error generates the pressing force. The desired orientation remains fixed at the measured contact-entry orientation. There is no commanded alignment trajectory.

Case A establishes the TCP-centred baseline for nominal zero, positive and negative entry tilt. Case B varies rotational stiffness about t_1, including five, fifteen and fifty newton metres per radian. Case C varies translational stiffness along t_2, using three hundred, eight hundred and two thousand newtons per metre. The t_1 translational stiffness remains fixed. Case D varies the tangential CoC displacement between minus forty and plus forty millimetres for both entry-tilt directions.

Within each comparison, the other settings remain unchanged. The separate null-space experiment is introduced next.

Null-space experiment

The second experiment holds a captured Cartesian end-effector pose and applies a repeatable commanded joint disturbance. The disturbance is the joint-torque equivalent of a virtual twenty-newton point force on link three. It is generated internally rather than applied by an external person or sensor.

$$\tau_{\text{dist}}(t) = J_p(q(t))^\top f_{\text{dist}}(t).$$

The force direction is selected to excite the redundant configuration. The Cartesian impedance opposes the disturbance's task component while maintaining the desired position and orientation. The measured acceptance criterion concerns TCP position retention.

$$\|p_d - p_{\text{EE}}\| < 2\,\text{mm}.$$

The pose-hold stiffnesses are two thousand newtons per metre for translation and fifty newton metres per radian for rotation, with fixed translational and rotational damping. The virtual disturbance is smoothly applied after the initial settling interval.

Four settings are compared in three trials each: no null-space torque, projected damping of two newton metre seconds per radian, and conditioning torques of one point five and two newton metres. The selected null-space law is active from the start, so conditioning may already move the configuration before the disturbance.

The results distinguish accumulated projected motion, which includes motion in both directions, from net projected displacement, where opposite motions can cancel. The largest measured TCP position error will be reported with the results.

Before the consistency checks, I will define the angular quantities used in the contact experiment.

Angular quantities

The measured angular offset and the contact response describe two different rotations. The measured angular offset, θ_{meas,t_1}, describes the tool orientation at contact entry relative to the configured surface reference. It is calculated from the measured end-effector orientation and the calibrated tool normal.

The Chapter Four sketch shows the configured reference in red, the contact-entry orientation in solid green, and the contact-end orientation in dashed blue. Both tool-face directions are reconstructed from measured end-effector orientations using the same calibrated tool normal. Their common origin compares orientations. It is not a contact point. The first surface tangent points out of the page, so positive rotation is anticlockwise by the right-hand rule. The curved arrow and plus sign above and to the right of the inset origin show positive rotation from positive t two towards the surface normal.

The outer arc gives the measured angular offset from the configured reference to the entry orientation. The inner arc gives the contact response, γ_{t_1}, from the measured end orientation back to the held entry orientation. The response therefore has the opposite sign to the start-to-end end-effector rotation. For rotation towards parallel alignment with the configured surface, it has the same sign as the measured angular offset.

$$\theta_{\mathrm{meas},t_1} = t_1^{\top}\theta_{\mathrm{meas}}, \qquad \gamma_{t_1} = t_1^{\top}\gamma.$$

The response calculation uses the measured end-effector orientations and the surface tangent. It does not require the calibrated tool normal. At contact entry, the captured orientation reference gives zero rotational controller error even when the measured angular offset is non-zero. At contact end, the response vector equals the rotational controller error.

$$R_{\mathrm{CE}} = R_{\mathrm{EE}}(t_{\mathrm{start}})R_{\mathrm{EE}}^{\top}(t_{\mathrm{end}}), \qquad \gamma = e_R(t_{\mathrm{end}}).$$

Neither quantity is an independent measurement of the physical tool–surface angle. The response magnitude alone does not establish the remaining physical alignment error.

With these definitions, I will first check whether the controller quantities behave consistently, beginning with quasi-static plausibility.

Normal-force plausibility assessment

The consistency checks begin with the quasi-static normal-force assessment. It compares the implemented translational impedance with its quasi-static spring prediction. The controller held a captured reference, with the compliance centre at the TCP and the null-space torque inactive. A manual displacement was then applied along the surface normal.

In the equation, $K_{p,n}$ is normal translational stiffness. e_p is the position error, and $e_{p,0}$ is its value before the perturbation. Their difference references the calculation to that starting state. Multiplication by $n_s^{\top}$ selects the normal component. The subscript qs means quasi-static prediction.

With a stiffness of one thousand newtons per metre and a measured normal displacement of minus nineteen point six five millimetres, the prediction is minus nineteen point six five newtons.

The commanded mean is minus nineteen point six four newtons, a difference of zero point zero five percent. The model-estimated mean is minus twenty-two point two nine two newtons, thirteen point five percent from the command.

The close spring-to-command agreement supports the plausibility of the implemented translational impedance relation.

$$F_{n,\mathrm{qs}} = K_{p,n}\, n_s^{\top}\left(e_p - e_{p,0}\right).$$

The rotational plausibility assessment follows the same comparison logic.

Moment plausibility assessment

The rotational assessment uses the same idea, but the tool is rotated about the first surface tangent while the normal load remains applied.

Here, K_{R,t_1} is rotational stiffness about that tangent. e_R is the orientation-error vector, and $e_{R,0}$ is the error before the rotational perturbation. Multiplication by $t_1^\top$ selects the component about the first tangent.

The starting reference is now the loaded stationary state. Subtracting it removes the moment already present under normal load, so the comparison isolates the response to the additional rotation.

The measured rotation is six point five six five degrees. With rotational stiffness of fifteen newton metres per radian, the spring prediction and the commanded mean are both one point seven one nine newton metres to the reported precision.

The model-estimated mean is one point eight five two newton metres, a difference of seven point seven five percent from the command. This supports the plausibility of the implemented rotational impedance relation.

$$M_{t_1,\mathrm{qs}} = K_{R,t_1}\, t_1^\top \left(e_R - e_{R,0}\right).$$

These checks support the implemented spring relations. I will now compare commanded and model-estimated wrench during contact.

Commanded and estimated wrench

This comparison examines the commanded and model-estimated wrench during one TCP-centred contact trial, with $r_c = 0$. Wrench means the combined force and moment. Here, F_n is the normal-force component, and M_{t_1} is the moment about the first surface tangent.

Both quantities are compared using the same surface axes and TCP reference point. The model-estimated signal is the interaction estimate available from the robot.

The shaded final interval is stationary. Over that interval, the estimated normal force differs from the commanded mean by two point three two percent, and the estimated moment differs by four point eight seven percent.

The transient curves, particularly for the moment, do not coincide throughout the complete motion. The percentages therefore refer to the stationary comparison, not the full time history.

This gives consistency evidence for the reported wrench levels. It is not an independent validation of sensor accuracy, because the estimate comes from the robot's model-based estimator.

Together, the quasi-static checks and the stationary wrench comparison establish that the reported controller quantities behave consistently before the main parameter comparisons. The model-based estimate is not an independent sensor measurement. I will begin the experimental results with the TCP-centred baseline.

Baseline contact response

Here, the centre of compliance is at the TCP. The horizontal axis gives the measured angular offset about the first surface tangent for all three entry conditions: zero point six nine degrees, plus nine point three one degrees, and minus nine point four one degrees.

The first condition has zero configured pre-contact offset, but its measured angular offset is zero point six nine degrees. This is the measured component about the first surface tangent.

The vertical axis gives the contact response. The bars show mean responses across three trials: minus zero point nine seven degrees, plus seven point five seven degrees, and minus ten point eight three degrees, respectively.

For the positive and negative offset conditions, the response has the same sign as the measured angular offset. Both correspond to end-effector rotation towards parallel alignment with the configured surface, although their response magnitudes differ. The nominal-zero condition produced a small negative response.

These results establish the TCP-centred baseline. They describe angular response and do not independently establish the final physical tool–surface angle.

I will now show how stiffness changes this response.

Effect of stiffness

The plot legends use the same measured angular-offset notation as the baseline.

The left plot shows the effect of rotational stiffness. K_{R,t_1} means the stiffness resisting rotation about the first surface tangent. These trials used the positive-tilt condition, with the other settings held constant.

Increasing this stiffness from five to fifty newton metres per radian reduced the response from approximately seven point six to two point seven degrees: a reduction of about sixty-four percent.

This agrees with the spring interpretation: greater rotational stiffness opposes more of the rotation produced by contact.

The right plot varies K_{p,t_2}, the translational stiffness along the second surface tangent. Rotation about t_1 can be accompanied by tangential motion along t_2, which is why this stiffness was selected. Lower stiffness gives less restoring force for the same tangential position error.

Changing it from three hundred to two thousand newtons per metre produced a response span of only three hundredths of a degree, or about zero point four percent of the baseline response.

Within the tested range, the alignment response was therefore much more sensitive to rotational stiffness than to tangential translational stiffness.

The next comparison changes the compliance-centre position while retaining the stiffness settings.

Effect of CoC position

The horizontal axis shows r_{c,t_2}, the centre-of-compliance displacement along the second surface tangent. The vertical axis remains the contact response about the first tangent. The curves are labelled by positive and negative measured angular offset. The error bars show one sample standard deviation across three trials.

For the positive-tilt condition, a displacement of plus forty millimetres produced approximately seven point nine degrees of response, compared with seven point six degrees at the TCP. Moving the centre to minus forty millimetres reduced the response to approximately zero point two degrees.

For the negative-tilt condition, the supporting side reverses. A negative displacement supports the negative response, while a positive displacement reduces its magnitude towards zero. Moving towards zero here means less rotation, not an improvement in alignment.

Relative to the TCP-centred case, the supporting side increased the response magnitude by about four percent. The opposing side reduced it by approximately ninety-eight to ninety-nine percent. The main effect was therefore the selection of a preferred rotational direction.

The time histories connect this directional effect to the interaction moment.

Contact response and interaction wrench

These three plots compare three centre positions for the positive-entry-tilt condition. Black is minus forty millimetres, red is the TCP-centred case, and blue is plus forty millimetres. The horizontal axis is time during Contact Establishment.

The left plot shows the contact response. In these representative traces, the supporting displacement reaches its approximately steady response at about two point three seconds, compared with about three point six seconds at the TCP. This is about one point three seconds earlier. This comparison describes the displayed traces, rather than a general timing result.

The middle plot shows the estimated normal force. The three traces settle at similar pressing levels. Their final-second means are approximately minus eighty-three, minus seventy-nine, and minus seventy-eight newtons at minus forty millimetres, the TCP, and plus forty millimetres, respectively. These are the rounded values listed below the plots.

The right plot shows the robot's estimated interaction moment about the first surface tangent, referenced to the TCP. This is an estimate of the physical interaction, whereas the earlier compliance-centre equation described the moment commanded by the controller. The estimate comes from the robot's model-based external-wrench estimator, rather than an independent force–torque sensor.

Reversing the CoC displacement reverses the interaction moment, while the final normal-force levels remain similar.

These observations connect the direction-dependent response to the interaction moment. I will now show the results of the separate commanded-disturbance experiment introduced earlier.

Null-space results

The first null-space result is panel a of the thesis figure. It compares the same commanded disturbance with no null-space torque, shown in black, and projected damping, shown in red.

The vertical axis is cumulative projected null-space motion, E_N, in degrees. This measure accumulates motion in both directions. The horizontal axis shows time. Each curve is the mean of three trials, and the shading shows one sample standard deviation.

Projected damping reduced the cumulative motion from seven point five nine six degrees to five point six eight seven degrees, a reduction of approximately twenty-five percent. The damping coefficient was two newton metre seconds per radian.

This result shows that damping suppresses redundant joint motion under the tested disturbance. The following panel evaluates conditioning together with Cartesian position retention.

Conditioning and Cartesian retention

Panel b of the thesis figure shows the conditioning and Cartesian-retention evidence for the same two conditioning settings. The horizontal axis shows conditioning torque magnitude. The black curve reports the change in the minimum singular value, with the scale shown at the upper left. The red curve reports the mean peak TCP position error on the right axis. The horizontal grey line is the two-millimetre position criterion.

Both settings increased the local conditioning indicator under the tested disturbance. Since the geometric Jacobian combines linear and angular rows, this singular value is interpreted as a relative indicator under the fixed Jacobian convention.

The mean peak position errors were zero point eight eight nine millimetres and zero point nine eight three millimetres for the one point five and two newton metre settings. Error bars show the trial variation. Across all twelve trials and all four settings, the largest measured TCP position error was one point three zero four millimetres, below the two-millimetre criterion.

Together with the preceding motion comparison, these results show that the secondary controller can modify the robot configuration while maintaining the tested Cartesian position tolerance. The third panel compares net displacement across all four controller settings.

Net null-space displacement

Panel c gives the net projected displacement for all four settings: no null-space torque, projected damping, conditioning at one point five newton metres, and conditioning at two newton metres. The bars show means across three trials and the error bars show one sample standard deviation.

The mean net displacement was seven point five one seven degrees without null-space torque and five point six zero nine degrees with damping. The two conditioning settings left near-zero mean net displacements of zero point zero one five degrees and minus zero point zero zero six degrees, respectively. The negative value for the stronger conditioning setting indicates motion opposite to the reference projection direction.

Net displacement differs from cumulative motion because opposite movements can cancel. Although the two conditioning settings left similarly small net displacement, their cumulative motions were zero point two eight three degrees and one point six one nine degrees. The stronger setting therefore produced about six times as much cumulative motion.

The conditioning term actively drives the configuration, and a small net displacement does not mean that the joints remained still. The largest measured TCP position error across all settings was one point three zero four millimetres, below the two-millimetre criterion.

I will now finish with the main conclusions and proposed future work.

Conclusion

The main result is passive alignment through Cartesian impedance. Contact produced rotation towards the configured surface while the desired orientation remained fixed. This was observed for both angular-offset directions with the CoC at the TCP. The measured response describes end-effector rotation rather than an independent measurement of the final physical tool–surface angle.

Higher rotational stiffness reduced contact rotation. Increasing stiffness about t_1 from five to fifty newton metres per radian reduced the measured contact rotation by sixty-three point nine percent. Higher rotational stiffness therefore opposed more of the rotation required for alignment.

Changing tangential stiffness had almost no effect on contact rotation. The experiment varied stiffness for translation along the second surface tangent. Across the tested settings, the measured contact rotation varied by only zero point zero three degrees, about zero point four percent of the

TCP-centred reference. The stiffness along the other surface tangent remained fixed, so these trials do not establish its separate effect.

The CoC displacement could support or oppose alignment. It introduced an additional TCP moment and favoured one rotational direction. A suitable displacement could accelerate the response, while the opposite displacement strongly suppressed it. At the TCP, $r_c = 0$, the added directional contribution vanishes. This makes the TCP a neutral reference for both directions while the ordinary rotational spring and damper remain active.

Damping reduced the motion caused by the commanded disturbance, while conditioning countered its effect in the tested direction. Projected damping reduced cumulative null-space motion by twenty-five point one percent. Conditioning left near-zero mean net displacement under the tested virtual disturbance. It remained an active configuration objective and could initiate motion. The stronger conditioning setting also produced more back-and-forth motion, so small net displacement did not mean that the joints remained still. The largest measured TCP position error was one point three zero four millimetres, below two millimetres.

The first future-work point is a more rigid tool mount, especially to reduce mechanical play about the second surface tangent. The present mount's mechanical play may contribute to the variability in that direction and to differences between tool-face rotation and measured end-effector rotation.

The second point is adaptive CoC selection and return to the TCP after alignment. A temporary displacement could support the rotation required by the measured entry tilt. Returning to the TCP would then remove the directional preference and retain compliant behaviour in both directions. This adaptive strategy was not tested in the fixed-CoC experiments.

The practical conclusion is to allow the contact rotation through suitable stiffness, use CoC displacement according to the required direction, and control redundant joint motion while preserving the Cartesian task. Thank you for your attention. I welcome your questions.

Contact response, normal force and moment

This full comparison shows the three centre positions for the positive-entry-tilt condition. The upper panel gives the angular response, the middle panel the estimated normal force, and the lower panel the estimated interaction moment about the first tangent at the TCP.

The final-second normal-force means are approximately minus eighty-two point nine, minus seventy-nine, and minus seventy-eight newtons at minus forty millimetres, the TCP, and plus forty millimetres. These levels are similar across the three positions, while the interaction moment reverses between the two displaced centres. In these representative traces, the supporting displacement reaches its approximately steady angular response about one point three seconds earlier than the TCP-centred case.

The force and moment are model-based estimates supplied by the robot. This comparison connects the directional angular response to the interaction moment under similar final normal pressing levels.

Contact demonstration

This video shows the robot picking up the tool, approaching the surface, and establishing contact. It then enters the pre-grinding hold.

During the hold, I manually change the surface orientation. Normal pressing remains active, and the compliant tool adapts to the changed surface. This demonstrates adaptation during the hold, rather than the fixed-condition contact-entry comparisons in the result plots.

Here, the centre of compliance is at the TCP. The surface contact moment drives the rotation. The controller's rotational spring and damper remain active, but the additional moment from a displaced compliance centre is zero. The desired orientation remains fixed.

The following demonstration shows the responses to a disturbance under the three null-space controller modes.

Disturbance demonstration

This video shows three responses to a disturbance while the Cartesian controller holds the tool pose.

First, no null-space torque is active, and the disturbance produces internal joint motion. Next, projected damping is active. Its torque opposes the null-space joint velocity and reduces the motion.

Finally, conditioning is active. This torque adjusts the joint configuration to improve its motion capability. It can generate internal motion while steering the configuration, so its role differs from damping.

Across the three settings, the focus is on how the arm configuration changes while the tool remains approximately in place. The preceding results distinguish accumulated motion from net displacement and show the Cartesian position-retention check.

This demonstration illustrates the distinct roles of null-space damping and conditioning.